

Pigmented Episcleral Deposits after Brachytherapy of Uveal Melanoma

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Purpose: To describe the characteristics and evolution of pigmented episcleral deposits after brachytherapy for uveal melanoma to determine their origin and association with melanoma-related mortality.

Design: Noncomparative case series.

Participants: Two hundred eleven patients (108 males, 103 females; median age, 61 years; range, 14–88 years) who were treated with a single ruthenium and iodine plaque therapy (median dose to tumor base, 475 Gy and 392 Gy, respectively) for a choroidal and ciliary body melanoma. Median tumor diameter and height were 12 mm and 5.5 mm, respectively. Eighty-eight patients were treated prospectively during the study.

Methods: The number and location of pigmented episcleral deposits were recorded under the slit lamp during each visit after brachytherapy. The association of the deposits with tumor characteristics and survival was analyzed with logistic regression and Kaplan-Meier analysis, respectively.

Main Outcome Measures: Number and location of episcleral deposits, melanoma-related mortality.

Results: The pigmented episcleral deposits ranged from black and brownish spots to slightly thickened patches. Most deposits appeared within the first 6 months after brachytherapy. By 1 year, 85% (95% confidence interval, 77–93) of eyes had at least 1 deposit (median, 6). The deposits increased in number until 7 years from irradiation, and decreased with increasing distance from tumor center. An association between the number of deposits at 1 and 2 years and subsequent melanoma-related mortality could not be confirmed ($P = 0.80$ and $P = 0.31$, respectively).

Conclusions: Pigmented macrophage-related episcleral deposits are found in most eyes with uveal melanoma after brachytherapy. Their association with plaque size and isotope rather than with tumor size suggests that radiation atrophy of retinal pigment epithelium and choroid in addition to tumor regression contributes to the formation of the deposits. Knowledge of their existence may save patients from unnecessary enucleation. *Ophthalmology* 2006;113:865–873 © 2006 by the American Academy of Ophthalmology.

In 1971, Boniuk and Hawkins¹ drew attention to transscleral migration of pigment after cryotherapy in an eye that was diagnosed to have an amelanotic uveal melanoma. The eye was enucleated because the pigmented episcleral deposits were thought to represent extrascleral extension of the intraocular tumor. Histopathologic analysis revealed, however, that the tumor was an astrocytoma and that the deposits were benign collections of melanin.

We have observed similar pigmented episcleral deposits after brachytherapy of uveal melanoma in most treated eyes both over and distant from the tumor base. Their origin and

role in the sequence of events leading to tumor regression have not been studied, in contrast to the impact of brachytherapy on vasculature^{2–5} and tumor destruction.^{6–8}

It has been suggested that brachytherapy modulates factors such as microvascular density⁵ and tumor-infiltrating macrophages.⁵ Moderate to high numbers of tumor-infiltrating macrophages are present in the majority of primarily enucleated and irradiated melanomas.^{3–5,9,10} Their presence in enucleated eyes is associated with higher melanoma-related mortality,⁹ but the role of macrophages in tumor regression and survival after brachytherapy is unknown.

Macrophages function as scavengers and producers of matrix metalloproteinases and of angiogenic and growth factors^{11–13} and modulate extracellular matrix and tumor invasiveness.^{11,14,15} When antigen is injected to the anterior chamber, macrophages migrate from the eye to the spleen and other lymphoid organs and act as antigen presenting cells, contributing to anterior chamber-associated immune deviation.^{16,17}

We investigated the evolution and origin of the pigmented episcleral deposits found after brachytherapy and their relationship to melanoma-related mortality. If these deposits were transsclerally migrating macrophages from the irradiated tumor, the number of episcleral deposits could reflect the number of tumor-infiltrating macrophages, in

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